Student Performance Predictor Report

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Student Performance Prediction Project Report

1. Introduction

In today's world it is very important for students to analysis their performance , how are they performing and progressing , even teachers have to identify which students need help , who is underperforming etc , while managing many students , This projects uses machine learning , linear regression and supervised learning to predict students performance by predicting their marks and pass and fail.

The main aim of this project is to predict marks of students , based on certain factors, and then use these predicted marks to decide whether the student is pass or fail. This helps in understanding how data can be used to support decision-making in education.

2. Problem Statement

The problem statement of this project is to predict a student's marks by considering few important inputs such as:

- Attendance
- Previous grades
- Study hours
- Class participation

After predicting marks these marks are used to predict whether a student is pass or fail , by checking if these marks are greater than 40 or not , if more than 40 , then it is pass or else fail.

3. Why This Problem Matters

Predicting students performance is important for both students and teachers and is a important real world problem , teachers can identify weak students while managing many students , and students can see how they are progressing and what should they change . This project is an example of how machine learning can be applied in education sectors.

4. Approach

The step by step approach of this project is given below

1. **Data Creation**: A small dataset was used which consisted features like attendance, previous grades, study hours, participation, and marks.

2. **Data Preparation**: This dataset was then transformed into pandas Dataframe , so that we can easily handle and process.
3. **Feature Selection**: The input features like (attendance, previous grades, study hours, participation) were kept separate from the target variable (marks).
4. **Train-Test Split**: The dataset was divided into training and testing sets to evaluate the model properly.
5. **Feature Scaling**: Input features were scaled by using standardscaler , so that all values are on similar range , this will help the model to learn better.
6. **Model Training**: A Linear Regression model was used to learn the relationship between the input features and the marks.
7. **Prediction**: Then this trained model was used to predict marks for the given data.
8. **Pass/Fail Classification**: If the predicted marks are greater than 40 then student is pass or else he is fail

5. Key Decisions

Important key decisions that I took during this project are :

- I used Linear Regression because it is simple and easy to understand and does the work effeciently.
- I choose 40 marks for the pass and fail criteria .
- Choosing attendance , study hours etc data, for datset.

6. Flowchart of the Project :-

Data Collection → Data Preprocessing → Feature Scaling → Model Training
→ Prediction → Pass/Fail Classification

7. Challenges Faced

- Understanding regression, training, and scaling for the first time which are machine learning concept.
- Learning how to use libraries such as pandas, numpy, and scikit-learn correctly.
- In code handling errors and fixing issues during implementation.
- For predicting marks deciding which features to include.

- Interpreting the model's output and making sense of the predictions.
- Managing data preprocessing steps like splitting and scaling properly.

8. Results

The result is that the project successfully predicts the expected marks of students and classifies the students as pass or fail based on these predicted marks.

Example output:
Predicted Marks: 63.12 -> Pass
Predicted Marks: 38.50 -> Fail

This model can be made more accurate by Giving larger datasets .

9. What I Learned

While doing this project I learnt the following thingd

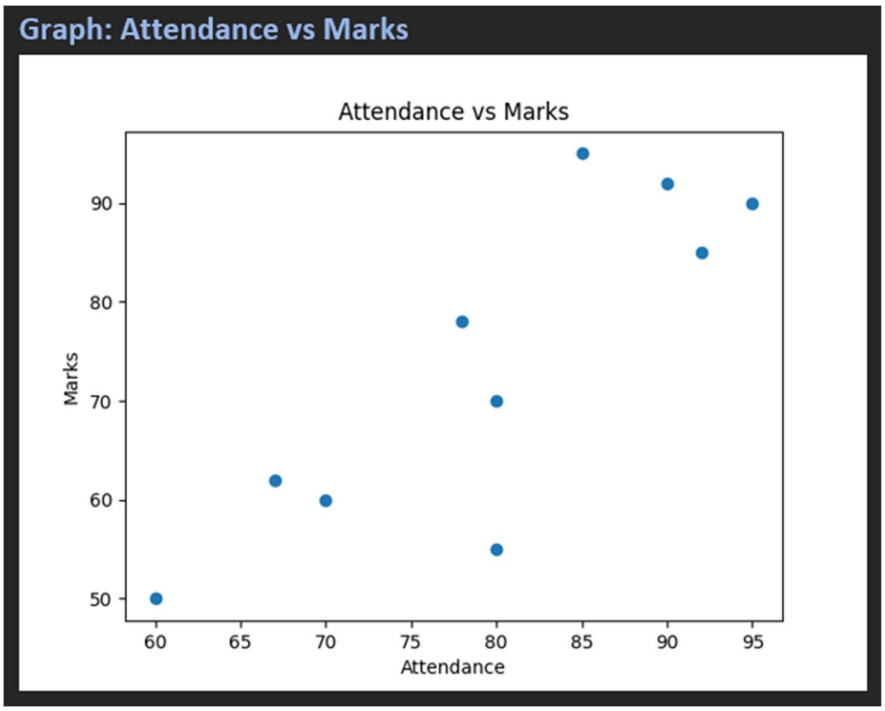
- I learnt the working of machine learning pipeline step by step
- I learnt the difference between regression (predicting marks) and classification (pass/fail).
- I learnt that preprocessing techniques like scaling plays an important role
- How useful insights can be provided by simple model

10. Future Improvements

There project can be improved in several ways in the future

- We can use larger and real world datasets for accuracy
- We can use advanced machine learning models like decision tree and random forest to improve this project
- We can add several types of graphs to understand the graph better
- For user interaction we can build a simple web or GUI interface.
- Model evaluation metrics can be included for deeper analysis.

11. Conclusion

This project shows a way to predict student performance using machine learning , it is about using data to make predictions , for example the model can predict if a student

will pass or fail . the model is not complex , it helps us understand how machine learning works in education , machine learning can also be used in areas.